GAUTAM BUDDHA UNIVERSITY

SCHOOL OF INFORMATION, TECHNOLOGY AND COMMUNICATION

HOUSE PRICE PREDICTION USING MACHINE LEARNING

Using Machine Learning & Linear Regression

Course	Machine Learning with Python
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Dataset	House Price Prediction Dataset

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House Price Prediction using Machine Learning

Abstract

The House Price Prediction project is a Machine Learning based web application developed to predict house prices using important features such as area and bedrooms. The system uses the Linear Regression algorithm for prediction.

Introduction

This project automates the process of predicting house prices using Machine Learning techniques integrated with Flask web development.

Objectives

- Understand Machine Learning concepts
- Implement Linear Regression
- Create predictive web application
- Integrate ML with Flask

Problem Statement

Predicting house prices manually is difficult because multiple factors influence the final value. This project solves this problem using Machine Learning.

Technologies Used

Python, Flask, scikit-learn, Pandas, NumPy, HTML/CSS, GitHub

Machine Learning Algorithm

Linear Regression is used to predict continuous numerical values based on input features.

Dataset Description

The dataset contains house information such as area, bedrooms, and price stored in house_data.csv.

System Architecture

User → Web Interface → Flask Backend → ML Model → Prediction Result

Features

- Real-time prediction
- Dark & Light Theme
- Responsive UI
- Beginner-friendly implementation

Advantages

- Fast predictions
- User-friendly interface
- Lightweight application

Future Enhancements

- Better UI/UX
- Cloud deployment
- Android App
- Advanced ML Algorithms

Conclusion

The project demonstrates practical implementation of Machine Learning integrated with Flask to create a functional house price prediction system.